

GR&R; Study Report

Gauge Repeatability & Reproducibility -- AIAG MSA 4th Edition | 10 Parts | 3 Operators | 3 Trials

Equipment / Gage:	Vernier Caliper VC-07 SN-1138
Study Performed By:	QE Team
Report Date:	2026-05-16
Study Design:	3 Operators, AIAG Crossed GR&R
Regulatory Reference:	21 CFR 820.72 AIAG MSA 4th Edition

Variation Components (%Study Variation)

Component	Study Variation	% of TV	Assessment
EV -- Equipment (Repeatability)	0.03808	60.1%	High
AV -- Appraiser (Reproducibility)	0.04243	66.9%	High
GR&R -- Combined	0.05701	89.9%	Unacceptable
PV -- Part Variation	0.02770	43.7%	--
TV -- Total Variation	0.06339	100.0%	--

Assessment column: GR&R; row uses AIAG pass/fail criteria ($\leq 10\%$ Acceptable / $10\text{-}30\%$ Marginal / $>30\%$ Unacceptable). EV and AV rows show diagnostic level (Low / Moderate / High) to identify the dominant variation source.

Number of Distinct Categories (NDC): 0 -- Inadequate: NDC should be ≥ 5 . Gage improvement required. [FAIL]

Intermediate Calculation Values

Parameter	Symbol	Value	Formula
Grand Mean	X-bar-bar	24.99563	Mean of all measurements
Avg Range (overall)	R-bar-bar	0.01249	Mean of per-operator R-bars
Upper Control Limit (R)	UCL_R	0.03214	$D4(2.574) \times R\text{-bar-bar}$
Operator Mean Spread	X-diff	0.01592	$\max(\text{op X-bar}) - \min(\text{op X-bar})$
Part Range	Rp	0.01710	$\max(\text{part avg}) - \min(\text{part avg})$
K1 constant (EV)	K1	3.05	$5.15 / d2^*(3 \text{ trials})$
K2 constant (AV)	K2	2.7	$5.15 / d2^*(3 \text{ operators})$
K3 constant (PV)	K3	1.62	$5.15 / d2^*(10 \text{ parts})$

Per-Operator Breakdown

Operator	Grand Mean (X-bar)	Avg Range (R-bar)
Alice	24.99504	0.01309
Bob	25.00389	0.01141

Operator	Grand Mean (X-bar)	Avg Range (R-bar)
Carol	24.98797	0.01296

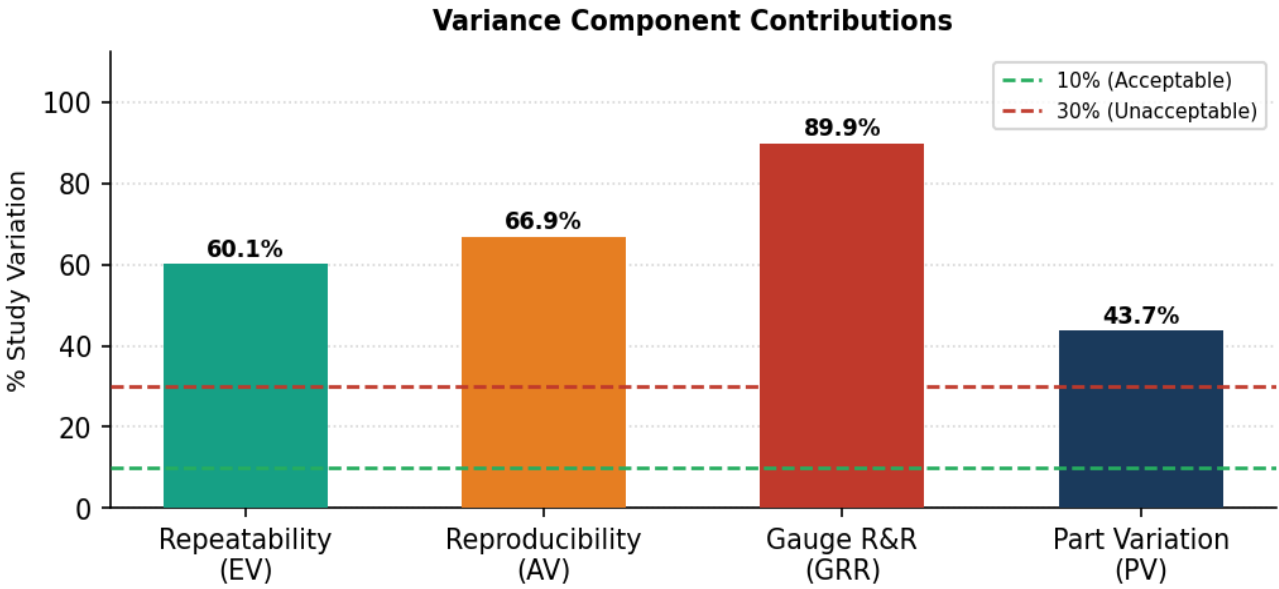
Acceptance Determination

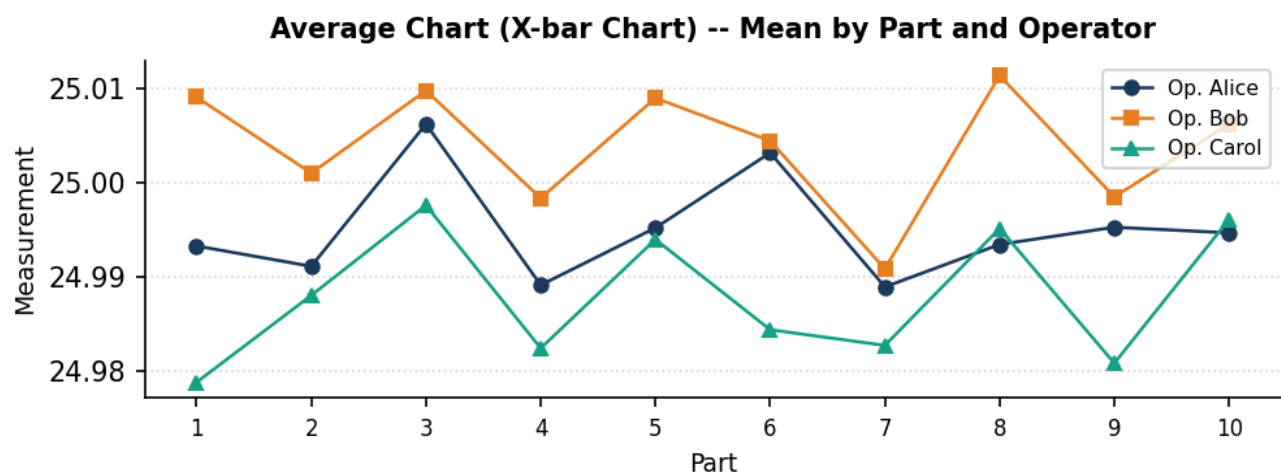
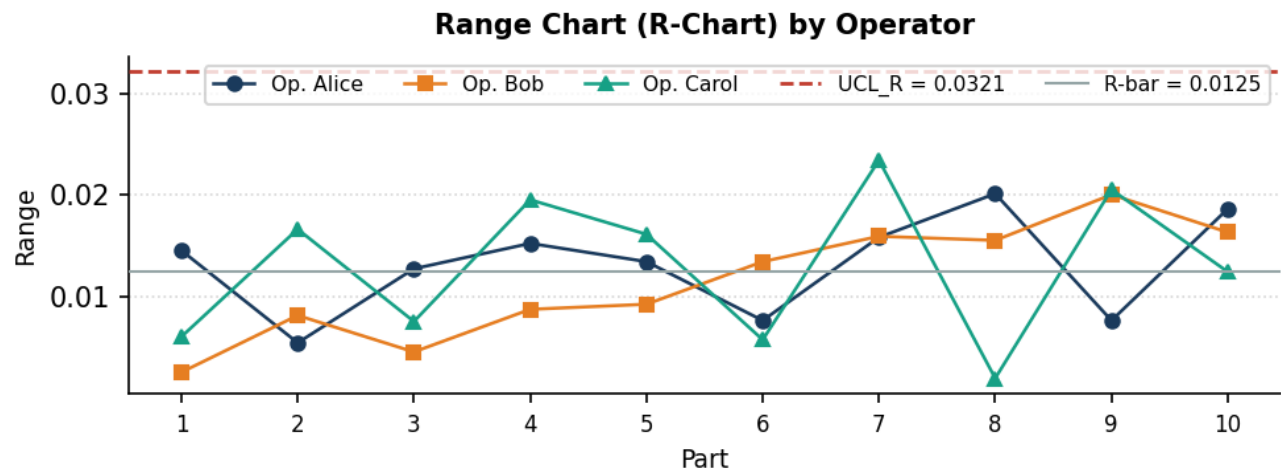
[FAIL] UNACCEPTABLE (%GRR = 89.9% | NDC = 0)

AIAG MSA Acceptance Criteria Reference

%GRR	Assessment	Recommended Action
<= 10%	ACCEPTABLE	Gauge approved. Measurement system is capable of product acceptance decisions.
10% - 30%	MARGINAL	May be acceptable depending on importance, cost, and customer concurrence. Investigate dominant variation source.
> 30%	UNACCEPTABLE	Do not use for product acceptance. Identify root cause and correct before re-study. Check gage discrimination, comp

Control Charts





The measurement system is UNACCEPTABLE (%GR&R; > 30%). Do not use this gage for product acceptance or process control. Identify and correct root cause before re-study. Common causes: insufficient gage discrimination, worn components, inconsistent measurement technique, or environmental factors.

Generated by `grr_tool.py` -- AIAG MSA 4th Edition Average & Range Method. Study variation expressed as 5.15-sigma (99% coverage). %Study Variation = $100 \times (\text{component} / \text{TV})$. Regulatory basis: 21 CFR Part 820.72 (FDA QSR). This report is a controlled quality record -- retain per document control procedures.